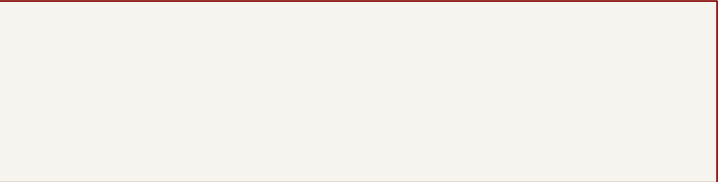


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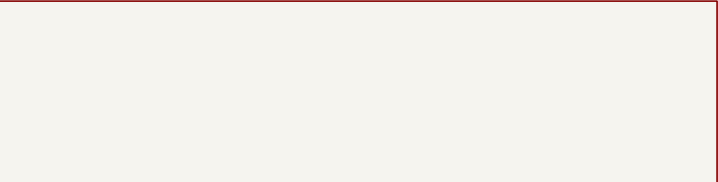
NIGHTWAVE AI—AUTHORED DIGITAL REFERENCE

Complete source package for independent engineering review; unbuilt and not approved for fabrication or lithium connection.
Power, charging and battery safety



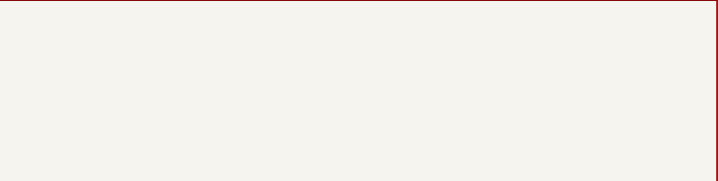
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MCU, USB and microSD storage



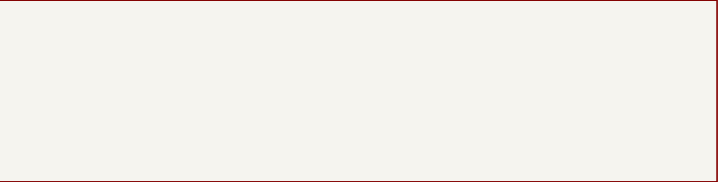
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I2S DAC, headphone and speaker audio



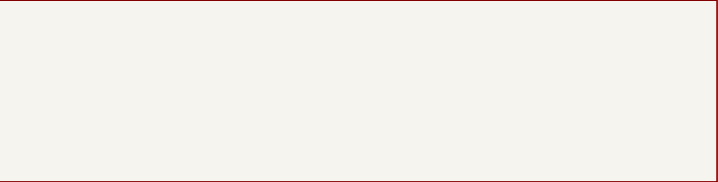
File: 03_local_audio.kicad_sch

TFT display, input expander and buttons



File: 04_display_controls.kicad_sch

BM83 Bluetooth source module



File: 05_bluetooth.kicad_sch

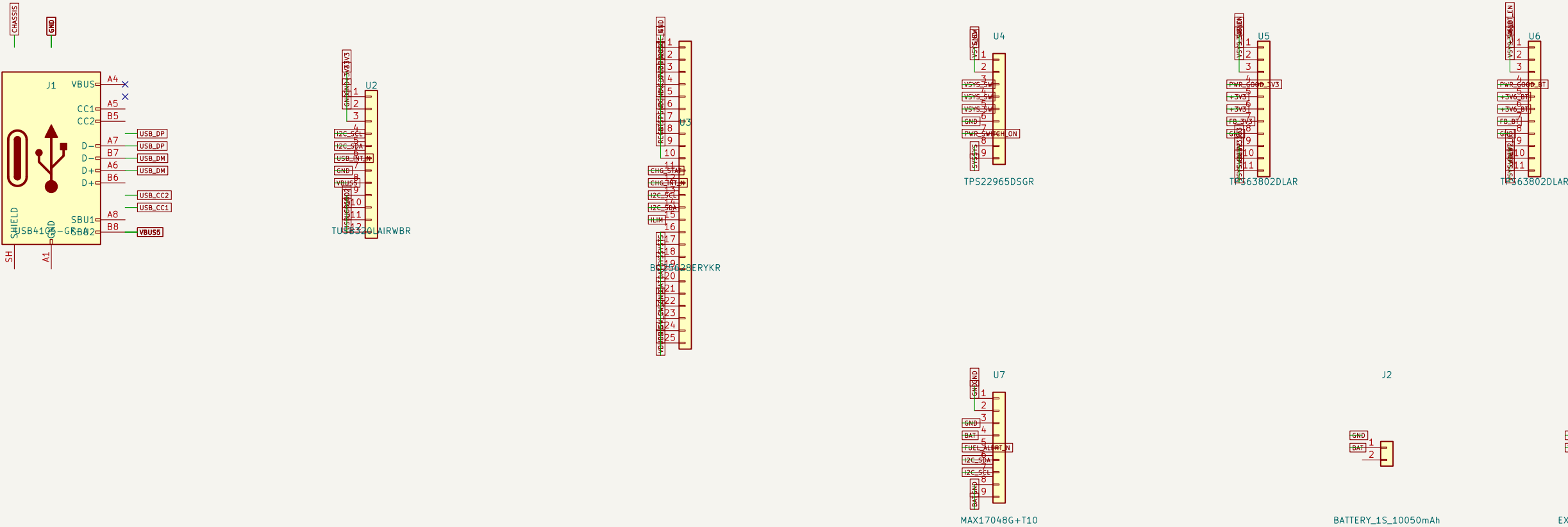
Service connectors, power switch and test points



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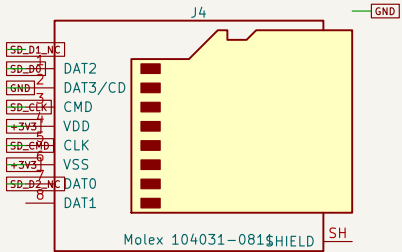
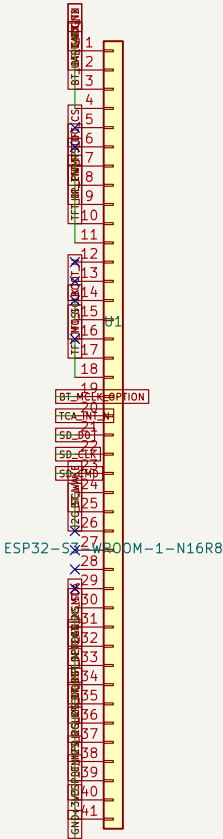
Power, charging and battery safety

AI-authored engineering reference. Verify pinout, support values, footprint orientation, lithium safety, RF keepout and assembly before fabrication.



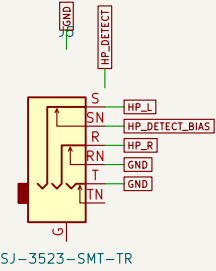
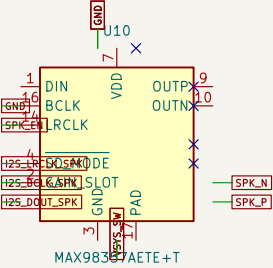
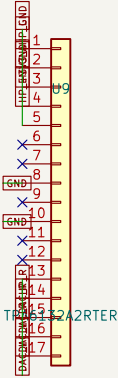
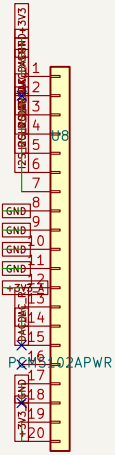
MCU, USB and microSD storage

AI-authored engineering reference. Verify pinout, support values, footprint orientation, lithium safety, RF keepout and assembly before fabrication.



I2S DAC, headphone and speaker audio

AI-authored engineering reference. Verify pinout, support values, footprint orientation, lithium safety, RF keepout and assembly before fabrication.



J6



CMS-28528N-L152



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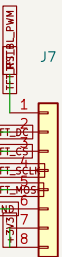
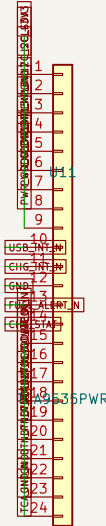
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TFT display, input expander and buttons

AI-authored engineering reference. Verify pinout, support values, footprint orientation, lithium safety, RF keepout and assembly before fabrication.



Waveshare 24382 TFT

SW1



PREV

SW2



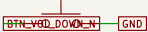
PLAY_PAUSE

SW3



NEXT

SW4



VOL_DOWN

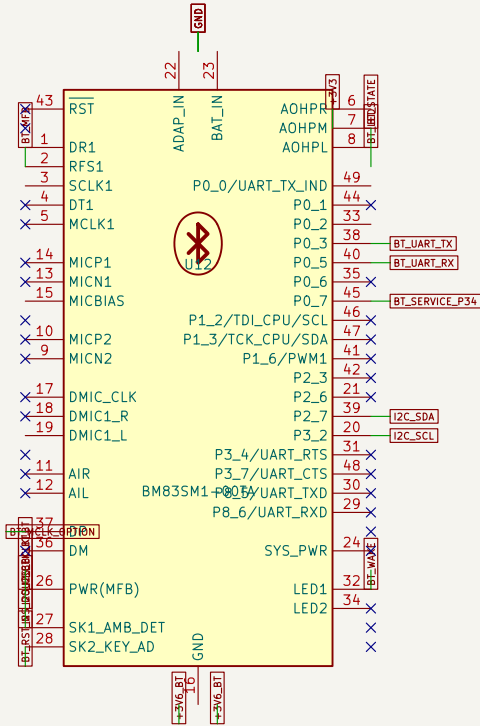
SW5



VOL_UP

BM83 Bluetooth source module

AI-authored engineering reference. Verify pinout, support values, footprint orientation, lithium safety, RF keepout and assembly before fabrication.



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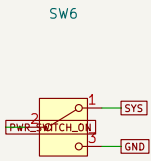


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Service connectors, power switch and test points

AI-authored engineering reference. Verify pinout, support values, footprint orientation, lithium safety, RF keepout and assembly before fabrication.



MAIN_POWER



UART0_SERVICE

TP1



VBUS5

TP2



BAT

TP3



SYS

TP4



VSYS_SW

TP5



+3V3

TP6



+3V6_BT

TP7



I2C_SDA

TP8



I2C_SCL

TP9



I2S_BCLK

TP10



I2S_LRCLK

TP11



I2S_DOUT

TP12



GND

Sheet: /Service connectors, power switch and test points/
File: 06_connectors_test.kicad_sch

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Date:

Rev:

KiCad E.D.A. 10.0.5

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